

Program: Diploma in Renewable Energy	
Course Code: 4291	Course Title: Electrical System Design & Estimation
Semester: 4	Credits: 4
Course Category: Program Core Course	
Periods per week: 4 (L:3, T:1, P:0)	Periods per semester: 60

Course Objectives:

- To make aware of acts and rules regulating the installation of electrical systems in India.
- To give basic knowledge of earthing and illumination technology.
- To provide fundamental knowledge in installation and electrification of residential and industry wiring.
- To understand basic concepts of solar PV wiring .

Course Prerequisites:

Topic	Course Code	Course name	Semester
Basic knowledge in Electrical & Electronics Engg.		Basics of Electrical & Electronics Engg.	1
Basic knowledge in Solar PV System.		Basics of Electrical Engg & Renewable Technology.	2

Course Outcomes:

On completion of the course, the students will be able to:-

CO _n	Description	Duration (Hours)	Cognitive Level
CO1	Identify the selection of suitable components for industrial/ non-industrial electrical installation.	16	Applying
CO2	Choose standard procedures for installation of lighting schemes.	14	Applying

CO3	Develop layouts and estimates for electrification of residential and industrial installations.	14	Applying
CO4	Develop design schemes for solar PV system installation.	14	Applying
	Series Test	2	

CO-PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3						
CO2	3						
CO3	3						
CO4	3						

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

On completion of the course, the students will be able to:-

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Identify the selection of suitable components for industrial/ non-industrial electrical installation		
M1.01	Choose relevant standards for electrification and standard format for estimation	2	Understanding
M1.02	Classify internal wiring circuits and components.	3	Understanding
M1.03	Select suitable components for power and light circuits in electrical installations.	5	Applying
M1.04	Prepare estimates for the electrification of domestic buildings.	6	Applying

Contents:

Relevant standards used for internal wiring - IS 732-various wiring methods as per IS 732- General rules for wiring - Relevant sections of CEA regulations regarding internal wiring. - review only.

Selection of cables for internal wiring-cable size calculation-simple problems. Selection criteria for control switches-main switch.-protective gears-RCCB-MCB-fuse. Coordination between fuses or MCBs at different circuit levels- size of earth continuity conductor and earthing conductor.

Estimation – definition – essential elements -specifications of various wiring accessories
Classification of Circuits-Light and Power circuits. Sub circuits- light sub circuit- power sub circuits. - Capacity of sub circuits.- determination of sub circuits -problems.
Switch boards-Main switchboard-sub switch board-Distribution board-Function – classification- specification.

Estimation of domestic(non-industrial) Installations -Estimation of electrification in domestic buildings- preparation of layout, wiring, diagram and estimates - house, office buildings

CO2	Choose standard procedures for installation of lighting schemes.		
M2.01	Explain the laws of illumination. and different terms related to illumination.	4	Understanding
M2.02	Select luminaires and lighting schemes for proper illumination at various places	3	Applying
M2.03	Design lighting scheme for domestic buildings	5	Understanding
M2.04	Choose appropriate schemes for street lighting	2	Applying
	Series Test – I	1	

Contents:

Relevant standards used for illumination - IS-3646 (1)-IS 6665 - review only

Illumination System -Define terms used in illumination – laws of illumination(statement & proof) -problems- various lighting schemes(Direct, indirect, semi direct and semi indirect.) –Recommended levels of illumination for various places(as per standards) – Space Height ratio – Utilization Factor - Depreciation Factor, Maintenance Factor- problems

Lamps and luminaries-classification-various types of light sources-incandescent lamps-fluorescent lamps-CFL-LED lamps - gas filled lamps - halogen lamps- arc lamps - -sodium vapour lamps. -Mercury vapour lamp. - neon lamp -luminous flux of various lamps- life span -applications-energy efficiency in lamps and illumination

Design of illumination schemes

Design of lighting schemes for domestic buildings--design considerations of good lighting schemes -calculation of number of lamps required- -problems

Industrial lighting- requirement-methods-street lighting-flood lighting

CO3	Develop layouts and estimates for electrification of residential and industrial installations.		
M3.01	Outline the rules regarding earthing systems	2	Understanding
M3.02	Prepare the estimate and cost of materials used for a standard earthing	2	Applying
M3.03	Prepare schematic diagram and estimates for the electrification of industrial installations	5	Applying
M3.04	Prepare schedule of electrical material for the installation of domestic and irrigation pump sets.	5	Applying

Contents:

Relevant standards for earthing - IS 3043- review only-IE rules regarding earthing systems

Earthing - purpose-classification-rules regarding earthing as per IS 3043 – estimation of materials used for a standard pipe earthing and plate earthing.

Estimation of Industrial Installations

Single and three phase motors- -need of starter- selection of starters-rating of back up fuse-rating of cables- rating of earthing conductor-Prepare estimate for the electrification of small workshops below 50kW connected load- layout- main switch board/ sub switch board-design-single line diagram for the power wiring.

Estimation of Pump Sets -Preparation of estimate and schematic diagram for the installation of domestic and irrigation pump sets

CO4	Develop design schemes for solar PV system installation.		
M4.01	Identify the components of a solar pv system.	3	Understanding
M4.02	Identify the specifications of a solar module	3	Understanding
M4.03	Develop the layout of grid connected solar PV systems	2	Understanding
M4.04	Design a standard solar PV system	6	Applying
	Series Test – II	1	

Contents :

Standards for Solar PV systems -NEC 690 (Brief review only)-General guidelines for installation of solar energy systems

Classification of PV systems- central power station system, distributed system-
.Basic components of grid-connected PV systems with and without batteries-
Types of solar batteries-Flooded type-Gel type-Absorbed Glass Mat (AGM) type

Specifications of a solar module- electrical & mechanical parameters-module energy output-Standard Test Conditions (STC)parameters-cell temperature- irradiance -air Mass

de rating factor of a solar cell-important factors considered for installing a solar panel -direction -tilt -shading -temperature-preventive maintenance of solar module

Layout of single phase and three phase grid connected solar PV systems (2KWp single phase system & 5KWp three phase system).

Stand alone PV systems- selection of location for installation- Important terms - Kilowatt -hours per square metre per day, Daily peak sun hours- Calculation of energy demand- Inverter and charge controller rating- Sizing of batteries- simple problems- Sizing of PV array- Sizing of cables- simple problems- Design of a solar PV system- Problems.

Text / Reference

T/R	Book Title/Author
T1	K. B. Raina, S. K. Bhattacharya, Electrical Design Estimating Costing, NEW AGE; Reprint edition (2010).
T2	J. B. Gupta, A Course in Electrical Installation Estimating and Costing, S.K. Kataria & Sons; Reprint 2013 edition (2013).
T3	B.H Khan ,Non-Conventional Energy Resources,Mc GrawHill Education (India) Third Edition
R1	National Electric Code, Bureau of Indian Standards publications, 2011.
R2	Relevant Indian Standard – specifications (IS – 732, IS – 746, IS – 3043, IS – 900), etc.
R3	M. K. Giridharan, Electrical Systems Design, , I K International Publishers, New Delhi, 2 nd edition, 2016

R4	S. L. Uppal, Electrical Wiring Estimating & Costing, Khanna Publishers, 2008
R5	N. Rajendran, Electrical System Design Data Book

Online Resources

Sl.No	Website Link
1	https://www.electricaltechnology.org/2019/10/components-solar-panel-system-installation.html
2	https://www.electricaltechnology.org/2020/09/series-parallel-and-series-parallel-connection-of-solar-panels.html
3	https://www.electricaltechnology.org/2014/06/solar-panel-wiring-installation-diagrams.html
4	http://ceikerala.gov.in/index.php/solar-pv-installations
5	https://www.electricaltechnology.org/2020/07/design-and-installation-of-solar-pv-system.html
6	https://www.energy.wsu.edu/documents/solarpvforbuildersoct2009.pdf
7	https://snv.org/assets/explore/download/standardised-training-manual.pdf
8	Solar PV design https://rdso.indianrailways.gov.in/works/uploads/File/Handbook%20on%20Installation%20&%20maintenance%20of%20Solar%20Panel(1).pdf